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III. *Dislocation Densities in Some Annealed and Cold-Worked Metals from Measurements on the X-ray Debye–Scherrer Spectrum*

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Abstract

Two basic equations are derived for deducing the dislocation density in powdered materials from the particle size and strain breadth measured from the Debye–Scherrer spectrum. In the particle size estimate, it is assumed that the material has a block structure similar to that found in microbeam studies and that the dislocations lie along the block surfaces. The number of dislocations along each face, n, is not known. In the strain broadening estimate the x-ray line broadening from a dislocation array is calculated in terms of the broadening due to an isolated dislocation and a strain energy factor F, which allows for the effect of dislocation arrangement. Both methods involve an unknown quantity but by equating the two results it is possible in most cases to get both a narrow bracket for the dislocation density and considerable information on the dislocation arrangement.

In annealed metals the values of ρ range from 2×10^7 cm of dislocation line per cm^3 for aluminium to 3×10^8 for tungsten and molybdenum, the majority of dislocations being ' random ', i.e. the number along each edge is approximately 1. There is some correlation between ρ and the purity and annealing treatment of the powder. High purity and high annealing temperatures result in smaller dislocation densities.

In cold-worked metals the dislocation density is more variable. A precise analysis is possible for aluminium ; high purity aluminium filed at room temperature gives $\rho=4{\cdot}5$ to $7{\cdot}6\times10^9$ with evidence for recovery, possibly by polygonization. Commercially pure aluminium filed at both room temperature and liquid air give $\rho=5\times10^9$ to $2{\cdot}7\times10^{10}$ and $\rho=1{\cdot}5\times10^{10}$ to $3{\cdot}2\times10^{11}$ respectively. Some dislocation array such as a ' pile-up ', which causes an increase in the strain energy of dislocations exists in both these samples as with all the other metals examined. Dislocation densities are also given for tungsten, molybdenum, iron and α-brass. In α-brass, the only alloy considered, ρ is higher than for the pure metals and there is strong evidence that the dislocations are effectively random and that each dislocation is attached to a stacking fault.

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§ 1. Introduction

Dislocation densities and information about dislocation arrangements have been deduced recently by Gay, Hirsch and Kelly (1953, 1954) from x-ray microbeam studies of cold-worked metals. The dislocation densities quoted are lower than previously accepted values and it appeared desirable to derive values from other x-ray results. In this paper values are deduced from x-ray measurements on the Debye–Scherrer spectrum of a number of metals and additional information about dislocation arrangements is given.

§ 2. Methods

Two quantities, both of which can be calculated from measurements on the Debye–Scherrer spectrum, may be used for calculating the dislocation density. The particle size may be derived from measurements of primary extinction (Williamson and Smallman 1955), line broadening (Williamson and Hall 1953, and Williamson and Smallman 1954) and from measurements of the peak shift produced by stacking faults (Paterson 1952 and Warren and Warekois 1953). The evidence from microbeam experiments indicates that the metal is broken up into blocks, the dislocations lying in the boundaries between the blocks. If there are n dislocations per block face the total length of dislocation line per block is $6\,n\,D/2$ (since each face is shared by 2 blocks) where D is the dimension of the block. The number of blocks per unit volume is $1/D^3$ and thus the dislocation density calculated from the particle size is

$$\rho = 6nD/2D^3 = 3n/D^2. \quad \ldots\ldots\ldots \quad (1)$$

To utilize this equation the value of n must be determined or assumed. The value $n=1$ gives a minimum dislocation density and could apply to annealed metals and to very severely deformed metals, in which state the dislocations may be very nearly random.

In deriving eqn. (1) we have defined ρ as the total length of dislocation line in unit volume. With this definition ρ is independent of the dislocation distribution. Densities expressed as the number of dislocation lines cutting unit area are usually equivalent, but an alternative interpretation of this has been employed which is not equivalent to eqn. (1). Gay, Hirsch and Kelly (1953), for example, have given values in terms of the effective numbers of dislocations cutting unit area with Burgers vector lying in the surface of the area. Using this definition one would obtain, instead of eqn. (1), that $\rho = n/D^2$, i.e. values $\frac{1}{3}$ of those quoted in the remainder of this paper*. This factor has been allowed for in comparing our values with those of Gay, Hirsch and Kelly in that the values quoted here are 3 times their published figures.

* The factor is $\frac{1}{3}$ only for an isotropic distribution of dislocations. For a non-isotropic distribution the latter definition is ambiguous and gives a variable value for ρ; by a special choice of the orientation of the area it is sometimes possible to make the two definitions agree.

From measurements of line profile it is possible to deduce the breadth of the strain distribution ξ (Williamson and Hall 1953). If $\overline{S^2}$ is the mean square strain of this distribution the stored energy of the lattice V is

$$V=3\,E\,\overline{S^2}/2=3\,E\,(A\xi^2)/2 \quad . \quad . \quad . \quad . \quad . \quad . \quad (2)$$

where E is Young's modulus and the factor A depends on the shape of the strain distribution. The energy of a dislocation v_c can be calculated in the absence of interaction with other dislocations and has been shown to be, for a screw dislocation

$$v_c=\frac{\mu \mathbf{b}^2}{4\pi}\ln\,(r/r_0) \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (3)$$

where **b** is the Burgers vector of the dislocation, μ is the shear modulus, r is the radius of the crystal containing the dislocation and r_0 is a suitably chosen integration limit, usually about 10^{-7} cm (see Cottrell 1954).

In reality dislocations interact so that their strain energies are changed on the average by a factor F, i.e.

$$v_{\text{real}}=v_cF. \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (4)$$

The dislocation density ρ calculated from the strain broadening is thus

$$\rho=\frac{V}{v_{\text{real}}}=\frac{k}{F}\cdot\frac{\xi^2}{\mathbf{b}^2} \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (5)$$

where $k=6\pi EA/\mu\,\ln\,(r/r_0)$. An approximate value of k can be found simply ; E/μ is fairly constant for the metals studied here and is approximately 2·6, $\ln\,(r/r_0)$ is a slowly varying function of r a reasonable value being 4, so that $k\sim 12\,A$. Williamson and Smallman (1954) have investigated the effect of line shape and have shown that for cold-worked metals the profile of the strain distribution lies between the extremes of a Cauchy and Gaussian distribution ; for the Cauchy $A\sim 2$, and for a Gaussian $A=\frac{1}{2}\pi$. Thus k lies in the range 2 to 25.

A more exact calculation of k which does not involve any of the above assumptions, is possible by using the results of Wilson (1949) for the broadening by a screw dislocation. This calculation is done in the Appendix and gives $k=16{\cdot}1$ for face-centred cubic materials with Burgers vector **b** along [110], and $k=14{\cdot}4$ for body-centred cubic metals with **b** along [111]. These values agree well with the approximate values deduced above and support the contention (Williamson and Smallman 1954) that the actual strain profile is very close to a Cauchy distribution.

Equation (5) can be applied directly to the results only if the interaction factor F is known or assumed. The simplest assumption (one that is always made in calculating dislocation densities from the calorimetric determination of stored energy) is that $F=1$. One model which approximates to $F=1$ is that in which one dislocation coincides with the edge of each block, the dislocations then have their maximum separation and interaction is a minimum. Evidently this model coincides with $n=1$ in addition to $F=1$ and it is easy to test the validity of these two assumptions since their correctness would imply that $\rho_p=\rho_s$, where ρ_p and ρ_s

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are the values of ρ calculated from eqns. (1) and (5) respectively in which values $n=1$ and $F=1$ have been used.

This possibility of testing from the experimental results whether a given dislocation arrangement is possible can be extended further and in addition to giving valuable information about the dislocation arrangement it considerably reduces the range of uncertainty in the final values for ρ. We shall consider two simple arrangements for which there is considerable experimental evidence and which appear to represent the two possible extremes for the factor F. If other models can be devised which produce a greater range of interaction factors these could be used equally well in the following discussion.

Piling up of the dislocations against barriers in the slip plane is commonly postulated as a mechanism for producing work hardening and this causes a large increase in the strain energy per dislocation. If there are n dislocations in each pile up they act to a first approximation as one dislocation of Burgers vector $n\mathbf{b}$ and hence the strain energy is increased from nv_c to n^2v_c (since it is the square of the Burgers vector which determines the energy in eqn. (3)). Thus $F=n$.

Polygonization on the other hand decreases the strain energy since the dislocations ' climb ' from their slip planes to form bend planes and their long range stress fields cancel at distances from the bend planes greater than h, where h is the separation of dislocations in the bend plane. The energy of each dislocation is reduced from the value given in eqn. (3) to the value corresponding to $r=h$. Thus

$$F=\ln (h/r_0)/\ln (r/r_0). \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (6)$$

However, $h=D/n$ and substituting eqn. (1) for n, $h=3/\rho D$, where ρ is the true dislocation density, also $r\sim\frac{1}{2}\sqrt{3}/\rho$ and $r_0=10^{-7}$ cm.*

Equation (5) now reduces to an equation for ρ, namely

$$\rho=\rho_s/F=\rho_s \ln\left(\frac{10^7\sqrt{3}}{2\sqrt{\rho}}\right)\Big/ \ln (3\times 10^7/\rho D). \quad . \quad . \quad . \quad . \quad (7)$$

This equation is best solved by trial and error starting with $\rho=\rho_s$ in the right hand side and, if the results are inconsistent with polygonization (i.e. F less than unity) the successive values of ρ rapidly diverge to infinity. If polygonization is shown to be impossible by this treatment a lower limit to ρ can be determined since pile up appears at present to represent the extreme degree of interaction. In this case $F=n$ so that $\rho=n\rho_p=\rho_s/n$ Thus

$$n=(\rho_s/\rho_p)^{1/2} \text{ and } \rho=(\rho_s\rho_p)^{1/2}. \quad . \quad . \quad . \quad . \quad . \quad (8)$$

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* This value of r_0 is used in calculating the energy per dislocation to exclude the dislocation core for within this region the energy is not proportional to the (strain)^2. If the value of h closely approaches r_0 a value $r_0=3\times 10^{-8}$ cm may be more appropriate since line broadening is sensitive to strain and not stored energy. In the case discussed in later sections h is much greater than r_0 and the difficulty does not arise.

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§ 3. Dislocation Densities in Annealed Metals

Annealed metals invariably exhibit extinction. Recent studies by Williamson and Smallman (1955 a) show that, for aluminium and α-brass, extinction is primary, and will probably be primary in all metals having a larger absorption coefficient and similar mosaic size. From measurements of the fall in intensity due to extinction it is possible to obtain the average linear dimension of the mosaic blocks D.

If a mosaic block is to reflect incoherently with its neighbouring blocks it must have a misregister of at least one atomic diameter along each edge (see Wilson 1949 a). Exactly such an effect would be produced by the block structure assumed in deriving eqn. (1) and the values of D deduced from measurements of extinction (or particle size broadening) do represent the physical quantity implied by D in eqn. (1). Confirmation of this argument, if it were necessary, is given by the very close correspondence between the values of D found in this study for aluminium and the values measured using microbeam methods (Gay, Hirsch and Kelly 1953).

Studies of extinction in a number of metals give the values of particle D shown in table 1. The value for tungsten has been derived from data for the 110 reflection only, the remaining values have been obtained using all the observed lines. Values of D for the annealed specimen of thoriated tungsten (Warren and McKeenham 1953) have not been included in table 1 since $D=8\times10^{-6}$ cm giving $\rho_p=4{\cdot}5\times10^{10}$; this sample is certainly not recrystallized. This result was anticipated because of the low annealing temperature used, 1050°C, compared to a temperature of 1350°C required for purer tungsten in the experiments of Williamson and Hall (1953).

The values for the strain broadening from annealed metals are more uncertain than the particle size. Williamson and Hall have shown that for aluminium and tungsten the slope of a plot of β^* against d^* has a minimum value (after correcting for the geometrical abberation of spectrometer) of about 3×10^{-4}. The maximum value of this slope is probably 4×10^{-4}. This broadening has been tentatively attributed to strains by Williamson and Hall (1953). However, a reliable estimate of the strain breadth is difficult since the natural breadth of the Kα doublet produces an effect indistinguishable from strain broadening For copper Kα radiation the half peak width of the wavelength distribution is 5×10^{-4} Å and the apparent strain breadth is $5\times10^{-4}/1{\cdot}5\simeq3\times10^{-4}$.

Since the maximum experimental error in this determination is approximately 1×10^{-4}, then the probable value for the strain breadth is in the region of $0{\cdot}5\times10^{-4}$. Taking **b** as $2{\cdot}7\times10^{-8}$ cm the value ρ_s (i.e. $F=1$) is about 5×10^{-7}. This is in very close agreement with the values of ρ_p given in table 1 and justifies the appliaction of the block model with one dislocation at each block edge in discussing the annealed state. Only the values ρ_p are given in table 1 as these are the more accurate.

An important correlation also appears between the variations in ρ and the annealing temperature in table 1. This is best seen by comparing

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Table 1. Minimum Values of the Dislocation Density in Annealed Metals from Extinction Studies

Source of original data	Metal	Purity	Anneal Temp. T	λ	D (cm)	$\rho_p = \frac{3}{D^2}$	$\frac{T_M}{T}$
Williamson & Hall 1953	Al	99·99%	500°c	Cu K	$3{\cdot}5\times10^{-4}$	$2{\cdot}4\times10^{7}$	1·21
Averbach & Warren 1949	α-brass	—	350°c	Cu K	$1{\cdot}25\times10^{-4}$	$2{\cdot}4\times10^{8}$	2·12
Williamson & Hall 1953	W	99·9+%	1350°c	Cu K	0·8 to $1{\cdot}2\times10^{-4}$	3×10^{8}	2·28
Williamson & Smallman 1955 b	Fe	Armco	675°c	Co K	$1{\cdot}2\times10^{-4}$	2×10^{8}	
	Fe	Carbon-Free	675°c	Mo K	2×10^{-4}	7×10^{7}	1·96
	Mo	commercially pure	1360°c	Cu K	1×10^{-4}	3×10^{8}	1·82
	Mo	commercially pure	1860°c	Mo K	2×10^{-4}	7×10^{7}	1·43

the value of ρ with the ratio T_M/T where T_M is the melting point (°K) and T the annealing temperature. Aluminium, with the lowest value of ρ observed in these studies has the lowest value T_M/T (1·21) whilst tungsten with a temperature ratio (2·28) gives a dislocation density of about 3×10^{-8}. The values for molybdenum illustrate this effect most clearly since both samples had the same purity and only differed in their annealing treatment. This case shows that ρ is a very steep function of T_M/T. No quantitative correlation using all the results is possible since the purity of the material appears to have an appreciable effect. The values for iron differ considerably even though both samples had the same annealing treatment; this difference must be ascribed to the difference in carbon content since one sample had been decarburized (other traces of impurities would be present to the same extent in both specimens). The amount of carbon in the armco iron before decarburization impeded the motion of dislocations at room temperature to a sufficient extent for a yield point to be observed, no such yield point being detected after decarburization. However, the magnitude of the effect of the carbon on the value of ρ is at first surprising for at the temperature of annealing the carbon atoms would have a very fast rate of diffusion. The effect might be explained if it were assumed that, during prolonged annealing after recrystallization the dislocations are in constant motion so that the mosaic block pattern is constantly changing.

§ 4. Dislocation Densities in Cold-worked Metals

Measurements of strain breadth can be made with comparative ease for all cold-worked metals but the values for particle size can be deduced with accuracy only for those metals which exhibit extinction even after cold working. For the remaining metals the particle size has been estimated from the line broadening but this answer is very much less accurate. In table 2 two columns each for D are given to indicate the reliability.

The results on aluminium are most useful since accurate particle sizes are known. For high purity aluminium the trial and error solution of eqn. (7) rapidly converges on a dislocation density $\rho=7{\cdot}6\times10^9$ corresponding to a strain reduction factor of 1·69, $n=162$ dislocations per surface and to an angle between neighbouring blocks $\alpha\simeq1°$. This tilt is in good agreement with microbeam results, as is the particle size D. The results are consistent with polygonization having occurred, as suggested earlier by Williamson and Hall (1953) from a study of Fourier coefficients of the broadened profiles; but it is interesting that, even in this high purity material which recovers readily (so that in some circumstances it recrystallizes at room temperature), the effect of recovery on the line broadening is slight, the density is only 70% greater than ρ_s. Since the effect of recovery is expected to be less for all the other materials studied the values of ρ_s given in table 2 can be taken as upper limits to ρ.

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Table 2. Calculation of Dislocation Density in Cold-Worked Metals from Particle Size and Strain Broadening. A random dislocation arrangement is not possible since these two values do not agree

Ref.*	Metal and Purity	Treatment and Temperature	Strain Breadth ξ	b cm $\times 10^8$	Max. Stored Energy cals/gm	ρ_s ($F=1$)	D (cm) Extinction	D (cm) Line Broadening	ρ_p ($n=1$)
W. & H.	Aluminium 99·99%	Filed 20°c	$4{\cdot}5\times10^{-4}$	2·85	0·004	$4{\cdot}0\times10^{9}$	$2{\cdot}6\times10^{-4}$		$4{\cdot}5\times10^{7}$
W. & H.	Aluminium (Commercial)	Filed 20°c	$1{\cdot}1\times10^{-3}$	2·85	0·03	$2{\cdot}4\times10^{10}$	2×10^{-4}		7×10^{7}
W. & S.	Aluminium (Commercial)	Filed −183°c	4×10^{-3}†	2·85	0·3	$3{\cdot}2\times10^{11}$	$0{\cdot}7\times10^{-4}$†		6×10^{8}
W. & H.	Tungsten (99·9%+)	Filed 20°c	$6{\cdot}5\times10^{-3}$	2·74	0·62	$7{\cdot}8\times10^{11}$		10^{-5}	3×10^{10}
W. & H.	Tungsten (99·9%+)	Annealed 1130°c	10^{-3}	2·74	0·015	$2{\cdot}1\times10^{10}$	$4{\cdot}5\times10^{-5}$		$1{\cdot}4\times10^{9}$
W. & S.	Molybdenum (Commercial)	Filed 20°c	$5{\cdot}5\times10^{-3}$	2·72	0·62	$5{\cdot}8\times10^{11}$		10^{-5}	3×10^{10}
W. & S.	Iron (Armco)	Filed −183°c	$5{\cdot}5\times10^{-3}$	2·47	0·60	$6{\cdot}9\times10^{11}$		10^{-5}	3×10^{10}
W. & S.	Iron (Armco)	Filed 20°c	4×10^{-3}	2·47	0·33	$3{\cdot}7\times10^{11}$		10^{-5}	3×10^{10}

* W. & H.=Williamson and Hall (1953), W. & S.=Williamson and Smallman 1955.

† The low temperature value for aluminium is based on the rapid measurement of one line, the breadth of which decreased even during this short time. The value of D is an estimate from the above value (see text).

This postulate is confirmed by an analysis of the results for commercially pure aluminium. Equation (7) has no solution unless $h<r_0$ and $\alpha>20°$. Tilts of this magnitude have not been observed in microbeam studies and polygonization is very unlikely. Thus $\rho_s=2{\cdot}4\times10^{10}$ appears to be an upper limit to the dislocation density in this case. The lower limit corresponding to severe pile up is $\rho=1{\cdot}4\times10^{9}$ (corresponding to $n=20$). Since commercially pure aluminium recovers some of its mechanical properties very readily at room temperature we would anticipate some 'unpiling' so that ρ would be in the range $1\text{–}2{\cdot}7\times10^{10}$. The results from the aluminium filed at liquid air temperatures are consistent with this, although a precise calculation is not possible as D is not known. A crude estimate of D is possible since Gay, Hirsch and Kelly (1954) have suggested that D is the slip band spacing, and Brown (1951) has shown that for deformation at liquid air temperatures the slip bands have one-third the separation they have at room temperature. If $D_{-183°}=0{\cdot}7\times10^{-4}$ ($D_{20°}=2\times10^{-4}$ cm) then there can be no polygonization as eqn. (7) has no solution. Equation (8) gives a dislocation density of $1{\cdot}5\times10^{10}$ if pile up is severe, whilst the upper limit to the density (ρ_s) is $3{\cdot}2\times10^{11}$. We expect less 'unpiling' in this sample and $\rho\sim2\times10^{10}$ is consistent both with this and the 'unpiled' value for the sample filed at room temperature.

A similar treatment is possible for the results from tungsten annealed at 1150°C which Hall and Williamson (1953) showed not to be recrystallized. Solution of eqn. (7) indicates slight recovery, the dislocation density being $2{\cdot}3\times10^{10}$, corresponding to a reduction in the strain energy per dislocation by 25%, with $n=20$; and to a tilt α of about 50′ of arc. A similar analysis could be done using the results from thoriated tungsten published by Warren and McKeenhan (1953) since their 'annealed' sample showed extinction corresponding to $D=8\times10^{-6}$ cm and $\rho_p=4{\cdot}5\times10^{10}$. It seems likely that this sample is polygonized and not truly annealed since the annealing temperature was 100°C lower than that used for the recovery treatment on the pure tungsten. The strain broadening was not measured although the strain breadth must exceed $1{\cdot}5\times10^{-3}$. Assuming by analogy that $n\sim10$ then $\rho\sim4{\cdot}5\times10^{11}$.

It is not possible to carry out such a detailed discussion for the remaining results listed in table 2 since exact values of D are not available. It is possible to set a lower limit to the particle size by line broadening studies and generally 10^{-4} cm $>D>10^{-5}$ cm. From the foregoing discussion on aluminium we do not expect any polygonization, though even if it does occur the effect on the upper limit of the dislocation density will be negligible. The range of values of ρ produced by taking these limits for D are given in table 4 for tungsten, molybdenum, iron filed at −183°C and iron filed at 20°C respectively.

Values for ρ_s can also be deduced by the Fourier analysis of line shapes which usually gives values of the mean square strain instead of the line breadth. The mean square strain for heavily cold-worked metals assuming a Gaussian profile is usually one-quarter that obtained assuming

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a Cauchy profile. The stored energy approach is used to give the two extreme values of ρ listed in table 3. Values for brass are taken from Warren and Warekois (1953), who only give values for the Gaussian profile, the upper limit is a very approximate estimate from this result.

Table 3. Calculation of Dislocation Density from the Mean Square Strain Determined by Fourier Analysis

Metal	Gaussian Profile		Cauchy Profile	
	$\overline{S^2}$	ρ_s	$\overline{S^2}$	ρ_s
Molybdenum	2×10^{-5}	3×10^{11}	8×10^{-5}	$1{\cdot}3\times10^{12}$
Iron	1×10^{-5}	$1{\cdot}9\times10^{11}$	6×10^{-5}	$1{\cdot}2\times10^{12}$
α-brass	6×10^{-5}	$1{\cdot}2\times10^{12}$	20×10^{-5}	3×10^{12}

The Gaussian and Cauchy profiles bracket the true value of S^2 and hence ρ_s lies between the two values given.

It should be noted that for α-brass ρ_s is larger than for pure metals. However, Warren and Warekois (1953) have shown that there are extensive stacking faults in this material and these, in addition to contributing to the broadening will act as barriers, maintaining any pile up. For 70/30 brass they give the separation of stacking faults to be approximately 120 Å and, assuming this to be the particle size, $\rho_p=2{\cdot}5\times10^{12}$ suggesting that $n=1$ and that virtually all dislocations are split into partials separated by an extended stacking fault.

§ 5. Discussion

For the metals examined in the present study the minimum dislocation density in annealed metals varies from $2{\cdot}7\times10^7$ to 3×10^8 and these values agree well with the values found using microbeam methods. The suggestion (made above) that high temperature anneals produce a lower dislocation density is supported by the work of Gay, Hirsch and Kelly who quote a value of $1{\cdot}5\times10^7$ cm of dislocation line per cm^3 for an aluminium crystal grown by a method which involves a prolonged anneal just below the melting point. The agreement between all these results is good; in annealed metals it appears that ρ can vary from 10^7 to 3×10^8 cm of line per cm^3, the value depending on both the annealing treatment and the purity. The dislocations in annealed metals are arranged in a fairly random fashion, each dislocation forming the edge of a mosaic block and producing the tilt necessary to make them reflect incoherently.

The probable range of values of dislocation density in cold-worked metals is listed in table 4. For heavily cold-worked pure metals the values are in good agreement with those of Gay, Hirsch and Kelly and are an order of magnitude or more lower than the accepted values. Brass, the only alloy considered here appears to have a dislocation density more

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nearly the conventional value, though this result appears to be associated with the presence of faulting.

In the last column of table 4 the type of dislocation arrangement is indicated as discussed in the text. The range of values for the dislocation density in the tungsten (filed 20°C), molybdenum (filed 20°C) and iron (filed 20°C and −183°C) correspond to the uncertainty in particle size which has been estimated from line broadening; ρ_{min} corresponds to $D=10^{-4}$ cm and ρ_{max} to $D=10^{-5}$ cm. It should be noted that there is no apparent reduction in the dislocation density in tungsten on giving it a recovery treatment at 1150°C sufficient to allow polygonization.

Table 4. Most Probable Range of Dislocation Density in Heavily Cold-Worked Metals

Metal	Treatment	ρ_{min}	ρ_{max}	Dislocation Arrangement
Al 99·99%	Filed 20°C	$4{\cdot}5\times10^{9}$	$7{\cdot}6\times10^{9}$	Polygonized
Al (99·7%)	Filed 20°C	5×10^{9}	$2{\cdot}4\times10^{10}$	Some pile up
Al (99·7%)	Filed −183°C	$1{\cdot}5\times10^{10}$	$3{\cdot}2\times10^{11}$	Some pile up
W (99·9+)	Filed 20°C	$1{\cdot}7\times10^{10}$	$1{\cdot}7\times10^{11}$	Some pile up
W (99·9+)	Annealed 1150°C	$2{\cdot}1\times10^{10}$	$2{\cdot}6\times10^{10}$	Polygonized
Mo (99·9)	Filed 20°C	$1{\cdot}5\times10^{10}$	$1{\cdot}5\times10^{11}$	Some pile up
Fe (99·85)	Filed 20°C	$1{\cdot}05\times10^{10}$	$1{\cdot}05\times10^{11}$	Some pile up
Fe (99·85)	Filed −183°C	$1{\cdot}6\times10^{10}$	$1{\cdot}6\times10^{11}$	Some pile up
Brass	Filed 20°C	2×10^{12}	3×10^{12}	Random

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APPENDIX

Calculation of Strain Broadening Produced by a Dislocation in the Powder Pattern

Wilson (1949, 1952) and Frank (1949) have considered the broadening produced by a screw dislocation. They find that the broadening is very anisotropic, each reciprocal lattice point being broadened in directions perpendicular to the Burgers vector of the dislocation but not in parallel directions. If the axes of reciprocal space are chosen so that the c^* direction is parallel to the Burgers vector then each spot is broadened to a halo whose breadth is given by

$$\beta^* = \frac{1}{\epsilon_0}(1 + 1{\cdot}33\, l)$$

where β^* is the reciprocal lattice breadth, ϵ_0 is the effective particle size of the crystal cylinder containing the dislocation and h, k and l are the orders of reflection along the a^*, b^* and c^* axes respectively. The component of the broadening due to strains is thus $1{\cdot}33\, l/\epsilon_0$ (the breadth of the strain distribution is reduced with increase in r, the radius of the cylinder since this increases the volume of material with low strain energy hence ϵ_0 appears in the term for the strain broadening). However this does not correspond to the broadening observed in practice for two reasons. Firstly the experimental broadening is sensitive only to the variation of the distance of reciprocal lattice intensity from the origin (i.e. the variation in d^*). Thus if ϕ is the angle between this line and the c^* direction the foreshortening factor is cosec ϕ (Wilson 1949). Secondly in a powder pattern on the average all possible reflections of each form will occur and these will not necessarily have the same value of l. Even in hypothetical primitive cubic crystals with dislocation axis along [001] it is clear that the 110, $\bar{1}10$, $1\bar{1}0$ and $\bar{1}\bar{1}0$ reflections have $l=0$, so that four out of the twelve (110) planes will make no contribution to the strain broadening. If ϕ is known, $l = d^*/\mathbf{b}^* \cos\phi$ where $\mathbf{b}^*$ is the reciprocal of the Burgers vector. Thus the effective strain broadening of each reciprocal lattice point is :

$$\beta_s^* = 1{\cdot}33\,(d^*\,\mathbf{b}^*) \sin\phi \cos\phi$$

and the strain breadth ψ (defined as β_s^*/d^*) of each point is

$$\psi = \frac{1{\cdot}33 \sin\phi \cos\phi}{\epsilon_0 \mathbf{b}^*} \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (6)$$

Wilson has calculated the effective particle size for a cylinder of radius, r and has shown that

$$\epsilon_0 = 16r/3\pi$$

and hence $$\psi = 0{\cdot}393 \sin 2\phi .\, b/r.$$

In table 5 the various values of ϕ, l, and their multiplicities p, ψ and $\bar{\psi}$ are given for possible reflecting planes from a face centred cubic lattice with Burgers vector along the $\langle 110 \rangle$ direction. $\bar{\psi}$ is the weighted mean defined as

$$\bar{\psi} = \Sigma p\psi / \Sigma p.$$

The mean of the breadth $\overline{\psi}$ averaged over lines 3, 4, 11, 12, 16, 19 and 20 is

$$\xi = 0{\cdot}244\ \mathbf{b}/r.$$

Since we are considering the case where there is no interaction between dislocations D^2 in eqn. (1) can be equated to πr^2 and thus

$$\rho = \frac{3}{\pi}\left(\frac{\xi}{0{\cdot}244\ \mathbf{b}}\right)^2 = 16{\cdot}1\xi^2/\mathbf{b}^2.$$

For body centred cubic metals with Burgers vector along the ⟨111⟩ direction $\xi = (0{\cdot}258)\ b/r$ and

$$\rho = \frac{3}{\pi}\left(\frac{\xi}{0{\cdot}258\ \mathbf{b}}\right)^2 = 14{\cdot}4\xi/\mathbf{b}^2.$$

Table 5. Calculation of the Strain Broadening Produced by a Dislocation in the Powder Pattern from Face-Centred Cubic Metals (Burgers Vector along ⟨110⟩).

Plane	ϕ	l	p	ψ	$\overline{\psi}$
111	35° 16	1	4	0·37 b/r	0·185 b/r
(and 222)	90°	0	4	0	
200	45°	1	4	0·393 b/r	0·262 b/r
(and 400)	90°	0	2	0	
220	0	2	2	0	0·227 b/r
	60°	1	8	0·34 b/r	
	90°	0	2	0	
311	31° 28	2	8	0·350 b/r	0·268 b/r
	64 46	1	12	0·303 b/r	
	90°	0	4	0	
331	13° 15	3	4	0·175 b/r	0·240 b/r
	49° 51	2	8	0·387 b/r	
	71° 41	1	8	0·241 b/r	
	90°	0	4	0	
420	18° 26	3	4	0·236 b/r	0·285 b/r
	50° 46	2	8	0·385 b/r	
	71° 34	1	12	0·236 b/r	
				$\xi =$	0·244 b/r

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